



June 5, 2026

SUBJECT: Independent Expert Evaluation and Recommendation for Somesh Rahul's Book, *The Anticipatory Home: A Blueprint for AI, Spatial Computing and Ambient Commerce*

To Whom It May Concern:

I am pleased to write this letter in support of Mr. Somesh Rahul and his book, *The Anticipatory Home: A Blueprint for AI, Spatial Computing and Ambient Commerce*. I have reviewed the book and consider it a meaningful contribution to the fields of human-centered artificial intelligence (AI), ambient intelligence, smart environments, spatial computing, and AI-enabled commerce.

By way of background, I am Vaibhav Unhelkar, Assistant Professor of Computer Science at Rice University, where I lead a research group in Human-Centered AI and Robotics. My research focuses on human-AI collaboration, with specific emphasis on human-robot interaction, interactive machine learning, and explainable AI. Concretely, my research develops AI-enabled systems that model human behavior, collaborate with people, and explain their decisions. I earned my Ph.D. in Autonomous Systems from the Massachusetts Institute of Technology, where my doctoral dissertation, *Effective Information Sharing for Human-Robot Collaboration*, examined how humans and robots can share complementary information to achieve fluent collaboration. I also earned a Master of Science in Aeronautical and Astronautical Engineering from MIT, and bachelor's and master's degrees in aerospace engineering from the Indian Institute of Technology, Bombay.

My scholarly work has been cited extensively in the fields of human-AI collaboration, human-robot interaction, and explainable AI. Based on my Google Scholar profile as of May 25, 2026, my work has received approximately 1,980 citations, with an h-index of 16 and an i10-index of 26. My highly cited publications include works on human-centered explainable AI, trust and transparency in automation, human-aware robotic assistants, bidirectional communication in sequential human-robot collaboration, and human-robot co-navigation using anticipatory indicators of human motion. At Rice University, I have taught advanced courses including Interactive Machine Learning (COMP 641), Reinforcement Learning (COMP 442/552), and Computational Human-Robot Interaction (COMP 565). These courses address topics such as human-in-the-loop machine learning, interpretability, reinforcement learning with human teachers, active learning, Bayesian inference, imitation learning, modeling human behavior from observational data, and sequential decision-making under uncertainty. This academic and technical background directly qualifies me to assess the significance of Mr. Rahul's book at the intersection of human-centered AI, anticipatory systems, intelligent environments, spatial computing, and responsible AI-enabled commerce.

Given my research and teaching in human-AI collaboration, explainable AI, interactive machine learning, and autonomous systems, I am particularly well positioned to evaluate whether Mr. Rahul's book offers a meaningful contribution to the design of AI systems that operate around people, infer human needs, explain their behavior, and preserve user agency.

Mr. Rahul's book addresses an important and timely problem: many systems marketed as "smart" remain rule-based, static, and burdensome for users. They automate isolated actions, but they do not reliably understand context, adapt to changing human needs, or earn user trust over time. The book argues that the next stage of smart homes and AI-enabled commerce must move beyond automation toward anticipation. In doing so, it contributes a practical and conceptually coherent framework for designing intelligent environments that serve people without removing their agency.

A central contribution of the book is the **Living Services Architecture**, or LSA, which Mr. Rahul presents as a four-loop framework: sensing, enhancement, visualization, and action. This framework is valuable because it gives builders of AI-enabled environments a clear order of operations. A system must first collect signals from the environment, then interpret those signals in context, then make its intended action visible or understandable to the user, and only then act. This framing is significant because many current smart systems collapse sensing and action into simple trigger-response rules. Mr. Rahul's approach correctly identifies that the missing layer is not merely better sensors or faster automation, but context-aware interpretation and trust-building interaction.

From a human-centered AI perspective, I find this framework especially valuable because it treats the user not as an obstacle to automation, but as the central participant in the design and implementation of the AI-enabled system. The book repeatedly emphasizes that anticipation must be earned through competence and explainability. These themes align closely with major concerns in research and practice of human-AI collaboration: how systems infer user intent, when systems should ask for confirmation, how systems should communicate uncertainty, and how users can retain control when intelligent systems act on their behalf.

The book's discussion of the **Trust and Control Plane** is another important contribution. Mr. Rahul recognizes that an anticipatory system cannot succeed merely by predicting correctly. It must also preserve user autonomy, provide understandable explanations, respect different consent levels across household members, and offer different levels of controls such as undo, pause, quiet mode, and full manual override. This is a sophisticated treatment of autonomy in domestic AI systems. It reflects a mature understanding that AI systems operating in private spaces must be evaluated not only by technical performance, but by whether users understand them, trust them, and feel safe living with them.

The book is also valuable because it connects multiple domains that are often treated separately: smart homes, AI product design, spatial computing, retail commerce, system architecture, and human-centered interaction. Mr. Rahul's professional background at Lutron and Walmart gives him a rare practitioner perspective across both physical home environments and large-scale digital commerce platforms. This combination allows him to articulate a vision of the "anticipatory home" not simply as a collection of connected devices, but as an adaptive service environment where AI, spatial visualization, ambient interfaces, and commerce can operate together responsibly.

In my opinion, the book's discussion of spatial computing is particularly relevant to the future of human-AI interaction. Mr. Rahul frames visualization not as a novelty feature, but as a trust-building mechanism. Before a system acts, especially in situations involving purchases, room changes, or irreversible decisions, the user should be able to preview what will happen. This "preview-commit" pattern is an important design principle because it reduces uncertainty, supports informed consent, and helps users build an accurate mental model of the AI system's behavior.

The book also introduces the **Anticipatory Readiness Index**, a maturity model intended to evaluate how far a system has progressed from simple connectivity to genuine anticipation. This is a useful contribution because the smart home and ambient computing industries lack a shared vocabulary for distinguishing basic connected devices from adaptive, context-aware, and trustworthy intelligent environments. By proposing measurable dimensions such as sensing capability, enhancement quality, action appropriateness, trust and transparency, and resilience, Mr. Rahul provides a structure that can support more rigorous discussion among engineers, designers, researchers, builders, and business leaders.

Based on my review, I consider this work suitable for academic use in advanced graduate-level teaching and research discussion, particularly in seminars addressing human-centered AI, interactive machine learning, human-AI collaboration, human-robot interaction, autonomous agents, and intelligent environments. In my teaching at Rice University, including Interactive Machine Learning (COMP 641), Computational Human-Robot Interaction (COMP 565), and Reinforcement Learning (COMP 442/552), I focus on how intelligent systems learn from human input, operate under uncertainty, support interpretability, and coordinate effectively with human users. Mr. Rahul's book is directly relevant to these themes because it translates abstract Human-Centered AI principles into concrete system-design patterns for real-world intelligent environments.

The sections on Anticipatory Readiness Index, Preview-Commit pattern, and Trust and Control Plane are particularly relevant for academic discussion because they connect technical AI concepts to deployed human-centered systems. They allow students to analyze questions that are central to human-centered AI deployed in practical contexts: when should an autonomous system act, when should it ask, how should it represent uncertainty, how should it learn from correction, how should it explain its behavior, and how should it protect user agency in a shared physical environment. In my opinion, this makes the book valuable not only as a practitioner blueprint, but also as a teaching and research discussion resource for advanced work in human-centered AI and intelligent interactive systems.

In my professional judgment, Mr. Rahul's book demonstrates originality in problem formulation and applied synthesis. While it builds on established ideas from human-centered design, ambient intelligence, smart environments, and AI-enabled decision support, it organizes them into a distinctive and practical blueprint for anticipatory living environments. The book does not merely summarize existing smart home concepts. It identifies a structural limitation in current systems, namely that many remain static and rule-based despite being marketed as intelligent, and proposes an integrated architecture for overcoming that limitation.

The importance of this work lies in its field-level relevance. The questions Mr. Rahul addresses are not narrow product questions. They are central questions for the next generation of AI systems: How should AI infer human intent? How should autonomous systems earn trust? How should intelligent environments explain themselves? How should commerce operate when embedded into private domestic life? How should users retain control when AI systems become increasingly proactive? These are important questions for computer science, human-centered AI, human-computer interaction (HCI), robotics, smart environments, and consumer technology.

For these reasons, I believe *The Anticipatory Home* is a valuable contribution to the emerging discussion around AI-enabled environments and human-centered intelligent systems. It offers a clear, forward-looking framework that can inform product leaders, engineers, researchers, builders, and students working on the future of smart homes, ambient intelligence, spatial computing, and responsible AI-enabled commerce. I offer this recommendation with confidence.

Sincerely,



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